

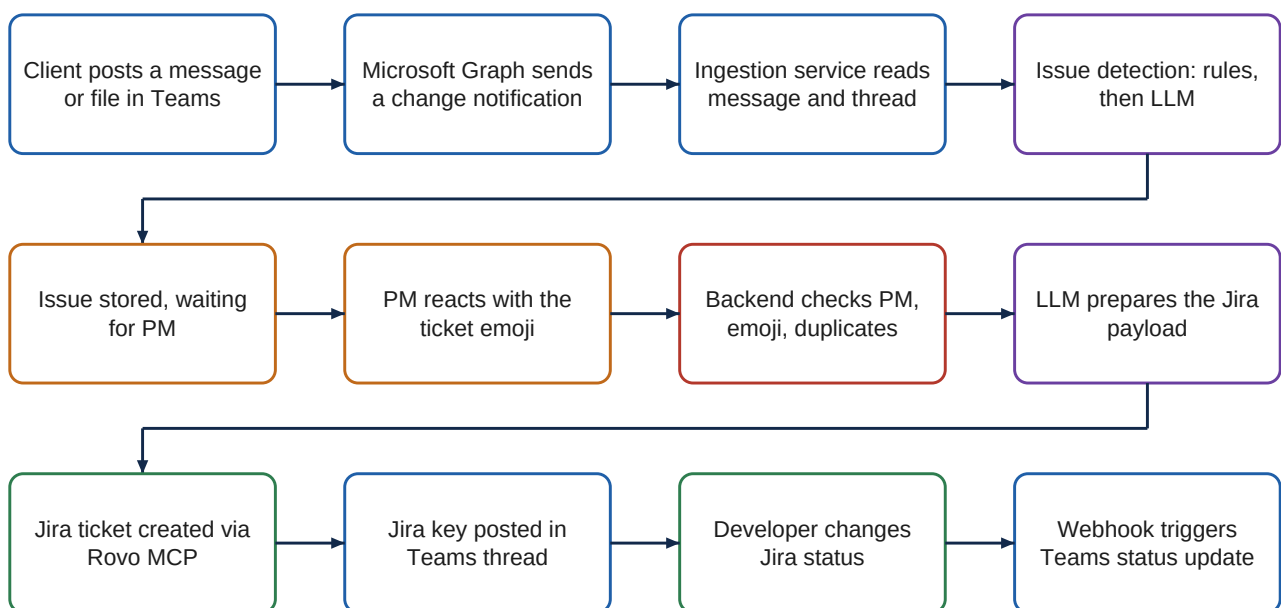
Teams to Jira Issue Automation

Implementation plan for client issue detection, PM approval, Jira ticketing and status sync

Purpose

Clients report problems in a Microsoft Teams channel or group chat. The system detects each issue, stores it, and waits for the Project Manager to approve by adding a configured emoji reaction. After approval it creates the Jira ticket through the Atlassian Rovo MCP Server, posts the Jira key into the same Teams thread, and keeps the thread updated as developers move the ticket forward. Clients follow everything in Teams and never need Jira access.

End-to-End Flow



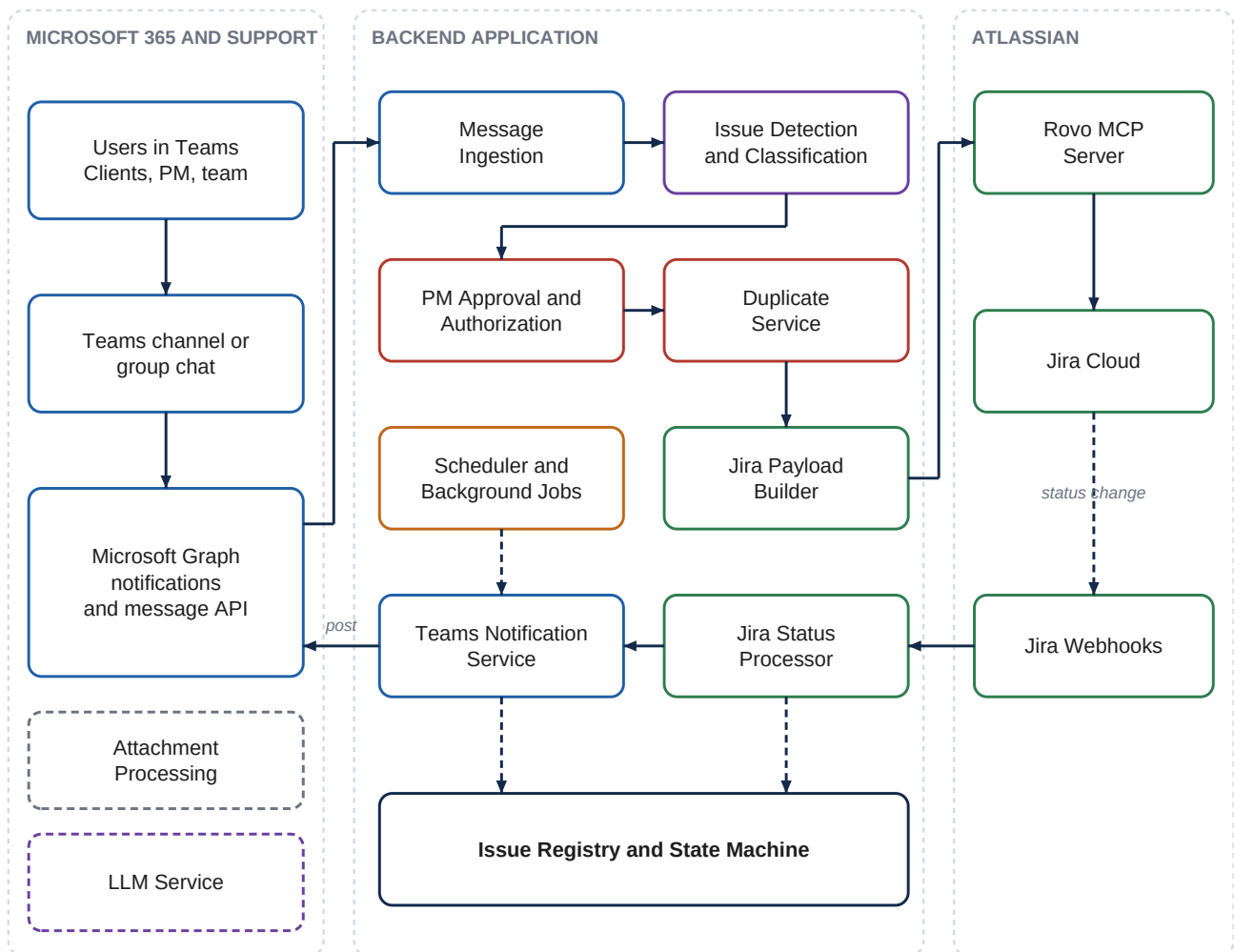
Blue: Teams side. Purple: AI-assisted work. Orange and red: approval and control. Green: Jira side.

Core Design Rules

- **AI classifies, the backend authorizes.** The LLM never approves or creates a ticket.
- **Approval is a validated reaction:** correct emoji, authorized PM (Entra user ID), approvable issue state.
- **One Teams message, one Jira ticket,** enforced by an idempotency key, a lock and a stored Jira key.
- **Configuration over inference:** project, Jira key, PMs and clients come from settings, not from the LLM.
- **Event-driven:** Graph notifications and Jira webhooks, with no continuous polling.

Architecture

One backend application with separated services. The Issue Registry is the single source of truth that links each Teams message to its Jira ticket.



Solid arrows: main flow. Dashed arrows: internal state access or asynchronous events.

Key components

Component	Responsibility
Teams app	Installed in each client team; enables Graph access and proactive messages
Issue Registry	Stores message IDs, state, approvals, Jira key, audit log and locks
LLM service	Classifies issues and drafts ticket content; no authority to create tickets
Jira MCP integration	Creates and updates work items, attachments and comments
Scheduler	Reminders, retries, subscription renewal and reconciliation

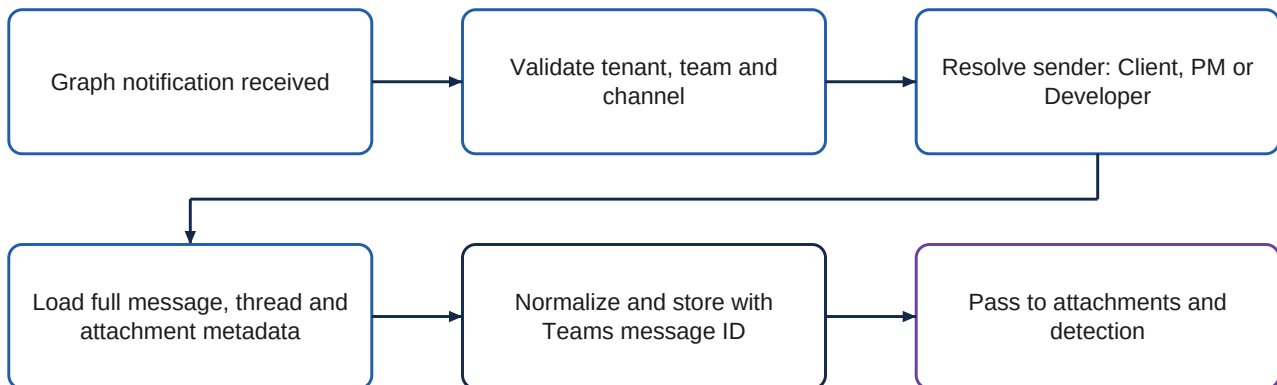
Configuration, Identity and Ingestion

Each Teams team or channel maps to one project record. The record fixes the Jira project, the PMs, the clients and the approval emoji, so none of it is guessed by the LLM.

Project configuration

Setting	Purpose
Tenant, team, channel or chat IDs	Which Teams conversation belongs to the project
Jira cloud ID and project key	Fixed destination for tickets
PM, client and internal user IDs	Role lookup by Entra ID; display names are never trusted
Ticket reaction and reminder interval	Approval emoji and minutes before a PM reminder (default 15)
Status mapping and attachment policy	Jira status to Teams message, allowed files and retention

Ingestion flow

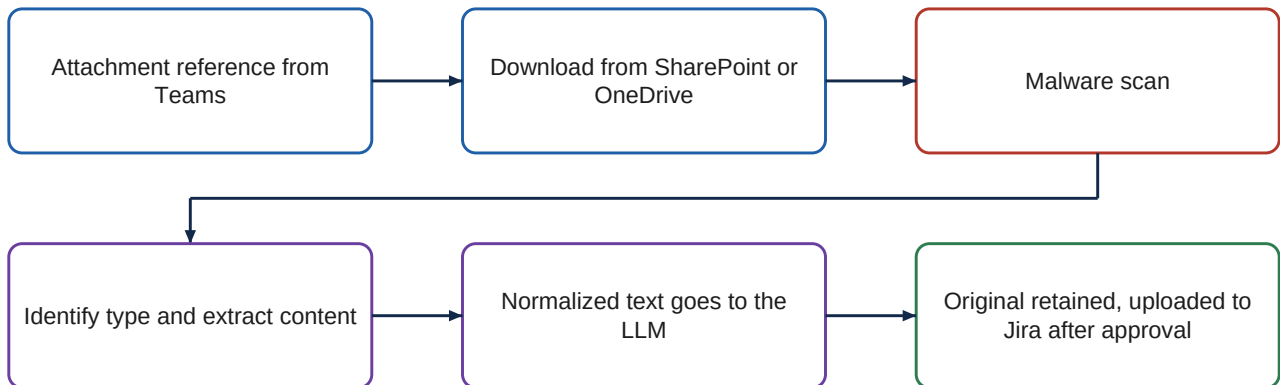


Graph subscription handling

- Acknowledge each notification at once and process it from a queue.
- Verify the client state value on every notification.
- Renew subscriptions before expiry and recreate them if a lifecycle event reports a gap.
- Edits update the stored issue; a deleted unapproved message marks the issue CANCELLED.

Attachment Processing

Clients may attach any file type. Files are never sent raw to the LLM: a dedicated service converts each one into normalized content and keeps the original under the retention policy.



Handling by file type

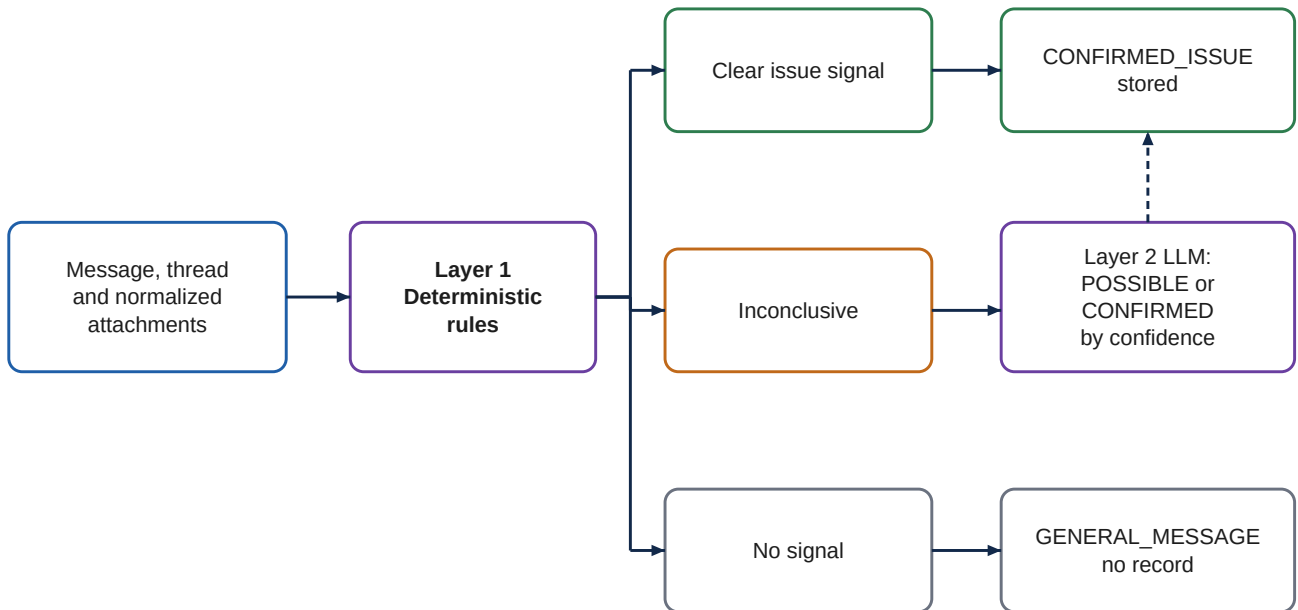
File type	Processing
PDF, DOCX	Text and table extraction, OCR fallback for scans
XLSX, CSV	Parser with headers and a bounded row sample
TXT and logs	Direct read with error code detection
Images	OCR and vision analysis
Video and audio	Transcription and key frames
ZIP	Controlled extraction with size and depth limits
Unknown or unsafe	Quarantine, keep original, metadata only

Rules

- Size, page, duration and archive limits are set per project; oversized files are flagged, not dropped.
- Extraction failures never block issue detection.
- Text found inside files is treated as untrusted and never followed as instructions.

Issue Detection and Classification

Detection never relies on one keyword such as "issue:". Two layers keep obvious cases fast and let the LLM handle ambiguous wording.



Layer 1: rules

Phrases such as issue:, bug:, error:, not working, unable to, failed, broken, cannot. Extra signals: a screenshot, log, video or document, an error code, or a request to fix a defect.

Layer 2: LLM structured output

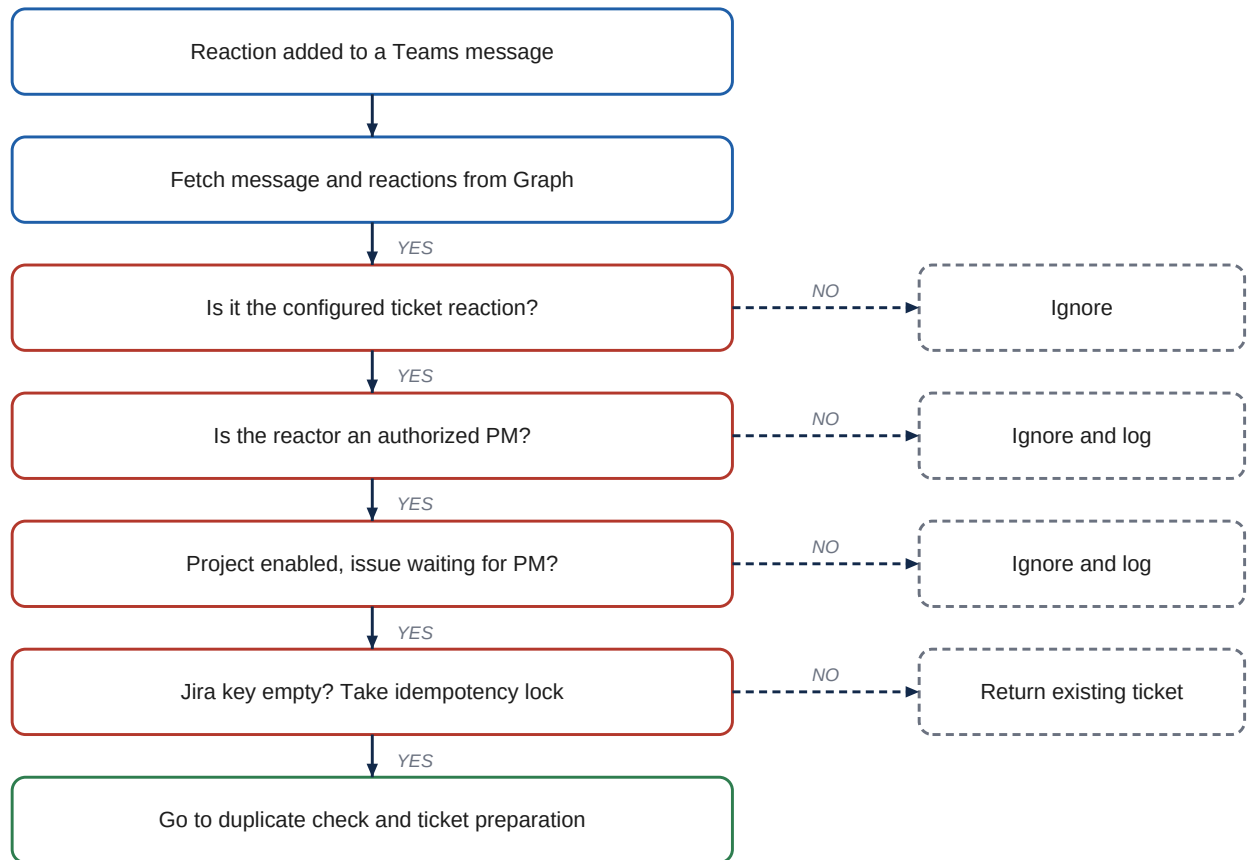
```
{ "is_issue": true, "confidence": 0.94, "issue_type": "Bug",  
  "summary": "Payment fails during checkout", "priority": "High",  
  "affected_module": "Payment", "evidence": ["Screenshot attached"] }
```

States

State	Meaning
GENERAL_MESSAGE	Normal conversation; no record
POSSIBLE_ISSUE	Below the confidence threshold; PM can still approve
CONFIRMED_ISSUE	Issue identified; waiting for PM, reminder timer starts

PM Approval, Authorization and Reminder

The reaction itself is the approval event. A fixed rule set runs on every Teams message update, and nothing in this path is decided by the LLM.



Authorization rule

```
if reaction == project.ticketReaction and reactor in project.authorizedPMs
  and issue.state in (WAITING_FOR_PM, REMINDER_SENT) and issue.jiraKey is NULL
then proceed, else do nothing
```

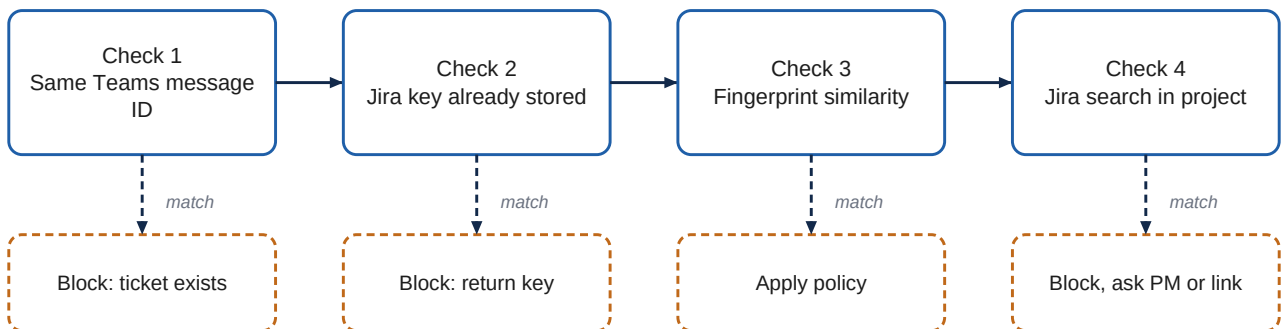
15-minute PM reminder



The reminder is posted as a reply to the original issue message.

Duplicate Prevention and Idempotency

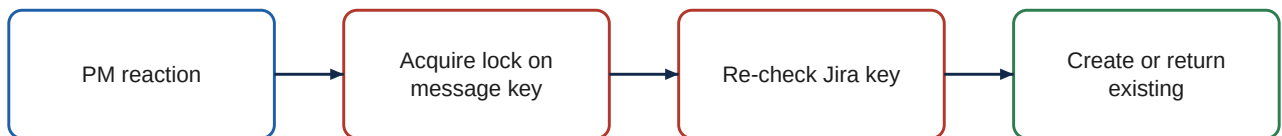
Four independent checks and a lock make sure one Teams issue always maps to exactly one Jira ticket, even with double reactions or webhook retries.



Policy (configurable per project)

Situation	Default behavior
Message already has a Jira key	Never create; reply with the existing key
High-similarity open ticket	Notify the PM with the key; PM links or forces creation
Client follow-up on known issue	Add a comment to the existing ticket
No similar issue	Continue to ticket preparation

Idempotency flow

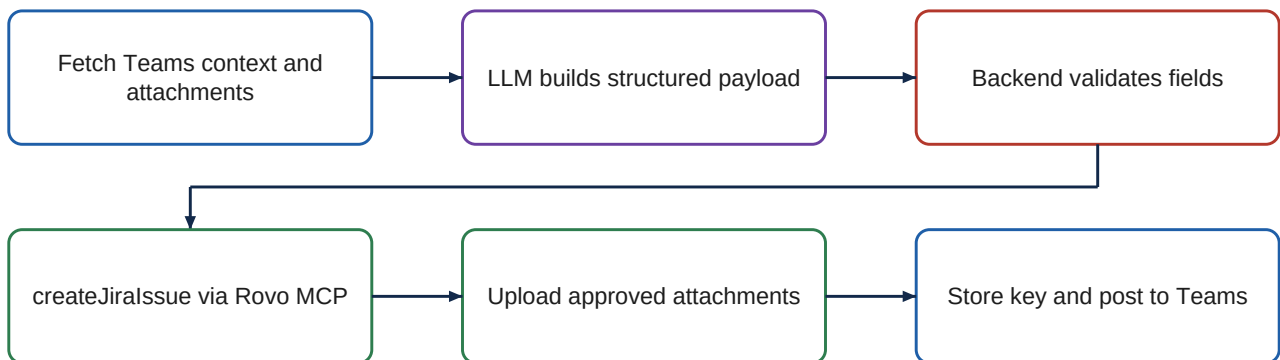


Edge cases

- Two PM reactions at once: the lock serializes them; the second returns the existing ticket.
- Crash after the Jira call: the CREATING state is reconciled by searching Jira for the Teams message URL.

Jira Ticket Preparation and Creation

The LLM drafts the content. The backend controls the project, the duplicate rules and whether creation is allowed at all.



Payload content

Field	Source
Summary, description, expected and actual result	LLM, written in Atlassian Document Format
Issue type, priority, module	LLM proposal checked against allowed project values
Project key, labels, custom fields	Project configuration only
Reporter, Teams message link, attachments	Teams sender identity, Graph data, approved files

Rovo MCP tools

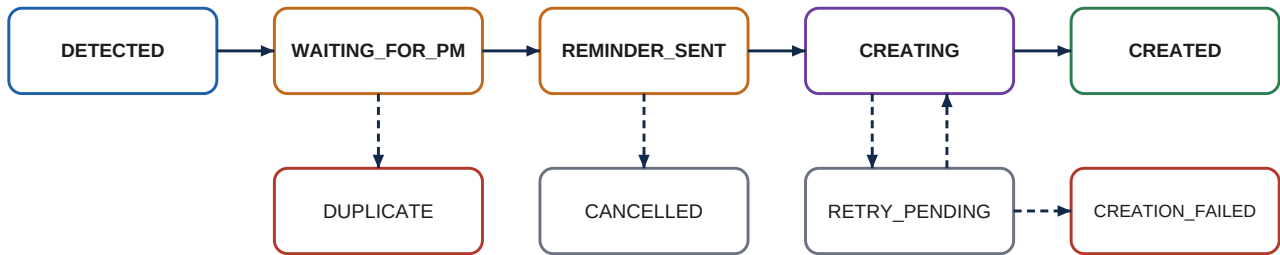
createJiraIssue, uploadAttachmentToJiraIssue, addOrEditJiraIssueComment, createJiraIssueLink, transitionJiraIssue and Jira search. The integration runs under a dedicated least-privilege service account.

Failure handling

Failure	Behavior
Jira timeout	RETRY_PENDING with backoff; reconcile by Jira search first
Invalid or missing field	CREATION_FAILED; PM told in the thread
Attachment upload fails	Ticket stays; upload retried separately

State Machine and Status Sync

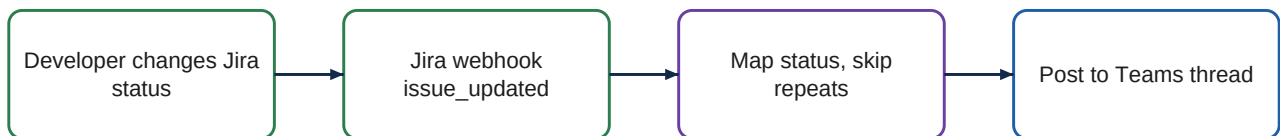
The Issue Registry holds one state per Teams issue. Every transition is logged with time and actor.



After CREATED, Jira synchronized states



Jira webhook flow



Jira webhooks replace polling. Duplicate events are dropped by event ID.

Status mapping (configurable per project)

Jira status	Message posted to the Teams thread
Open	Issue created, with Jira key, priority and links
In Progress	Development has started
Blocked	Development is blocked; PM mentioned
Testing	Fix is under testing
Done or Closed	Issue resolved, with Jira link

Data, Security and Operations

Issue Registry record

Group	Fields
Teams	tenant, team, channel or chat ID, message ID, thread ID, message URL, client user
Content	original message, normalized issue, type, priority, fingerprint, evidence
Approval	state, PM user, reaction time, reminder time
Jira	issue key, URL, status, created and resolved times

Security

Area	Control
Identity	Entra user IDs decide client and PM roles
Teams and Graph	Least-privilege permissions; client state validation
Jira and MCP	Service account with OAuth 2.1 or API token, limited projects
LLM	Approved provider, no tool access, strict output schema
Audit	Every approval, Jira call and notification logged

Reliability

- **Outbox pattern** for Teams messages and attachment uploads, with retry and backoff.
- **Scheduler jobs** for reminders, subscription renewal and reconciliation of unknown Jira states.
- **Idempotent queue handlers** so replays never create duplicate tickets or messages.

Dashboard

The first release needs only the database, Teams notifications and Jira links. A dashboard follows once the workflow is stable, showing issues by state, client, module and priority, plus average PM approval time and resolution time.